

02 — Learning Objectives (filled example)

Bloom-verbed, student-focused, measurable. Module objectives ladder up to the course objectives.

Course-level objectives

By the end of the course, students will be able to:

1. **(Apply)** Implement end-to-end ML pipelines from data loading to evaluation. — *Evidence:* a working notebook that runs load → preprocess → train → evaluate on a fresh dataset.
2. **(Evaluate)** Select appropriate algorithms based on problem type, data, and constraints. — *Evidence:* given a new problem and dataset, justify an algorithm choice against the constraints.
3. **(Create)** Write production-quality code using sklearn Pipelines and Keras training loops. — *Evidence:* submit a clean sklearn Pipeline / Keras loop that runs top to bottom and reads as production code.
4. **(Analyze)** Diagnose common ML failures: overfitting, data leakage, class imbalance. — *Evidence:* from a model's diagnostics, identify the failure and point to the evidence for it.
5. **(Apply)** Communicate model behavior and limitations to non-technical stakeholders. — *Evidence:* produce a plain-language explanation of a model's behavior and limits for a non-technical audience.

Week 1 module objectives (ladder up to CO1, CO4, CO5)

By the end of Week 1, students will be able to:

1. **(Apply)** Execute a complete ML pipeline: load → preprocess → train → evaluate → interpret. (→ CO1)
2. **(Apply)** Implement linear regression in sklearn with a proper train/test split. (→ CO1)
3. **(Interpret/Understand)** Interpret regression coefficients in business terms (“holding all else constant”). (→ CO5)
4. **(Analyze)** Identify residual-plot patterns for diagnostics (random scatter, funnel, curve). (→ CO4)
5. **(Evaluate)** Judge model fit using MSE/RMSE/R² and communicate results to stakeholders. (→ CO5)

Note: objective 3 leans on “interpret/understand” — acceptable here because stakeholder explanation is the measurable outcome (they produce the explanation).